Kgwebo

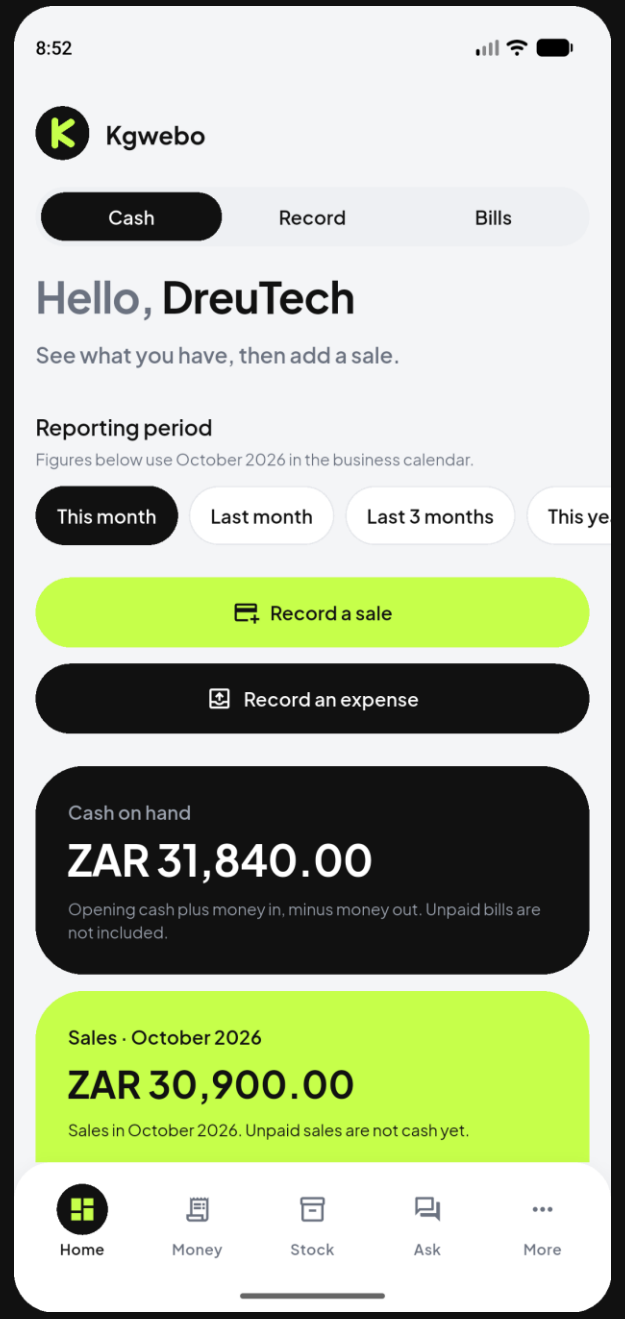
Helping small businesses understand their money.

A Flutter stall-finance app that records cash-basis activity, shows what cash is actually available, and turns those records into the next useful step.

2026 FNB App of the Year · Youth entrepreneurship & small-business growth

JUDGE DEMO ACCOUNT

Email: dreu.davids1@gmail.com
Password: password
Stall: DreuTech · illustrative records



THE PROBLEM

The till is busy. The picture of the money is not.

These are practical problems for informal traders and small service owners across Southern Africa.

Records live in notebooks and chat threads

Sales, expenses and stock are captured inconsistently, if at all.

Cash on hand is a guess

Money in the till is mixed with personal cash, unpaid sales and unpaid bills.

Upcoming bills arrive as surprises

Rent, software and supplier payments are not visible before they fall due.

Numbers rarely become a decision

A tracker that only lists transactions does not say what to do next.

THE SOLUTION

One place to record, understand, anticipate and act.

Kgwebo is a working Android app. It already records sales and expenses, keeps a stock catalogue, forecasts outgoing bills, and answers questions from recorded figures.

Record

Paid and unpaid sales, expenses, refunds, owner cash moves.

Understand

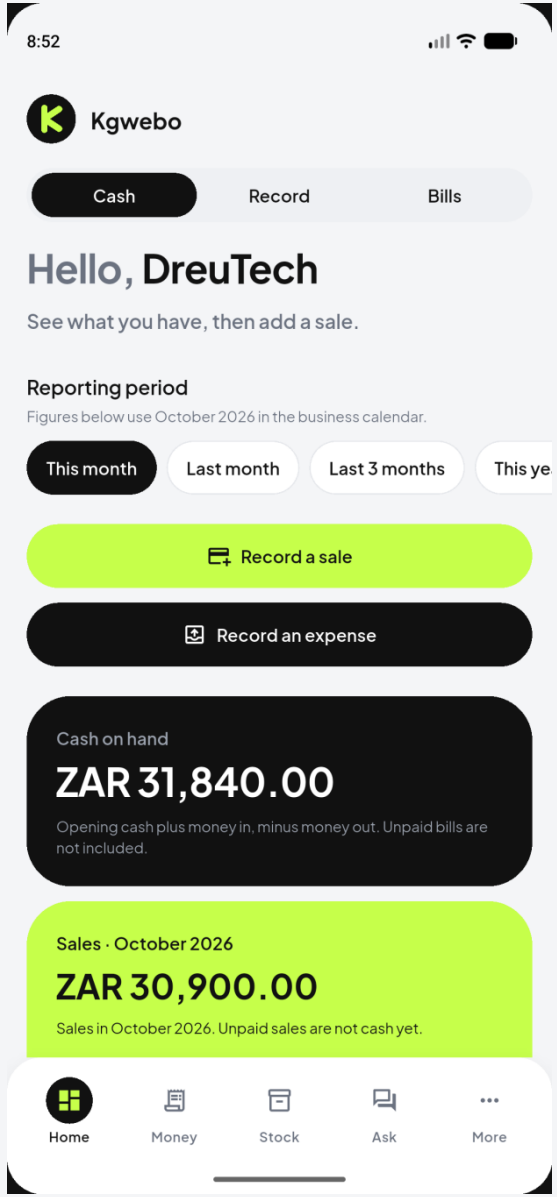
Recorded cash is separated from sales. Opening cash is not revenue.

Anticipate

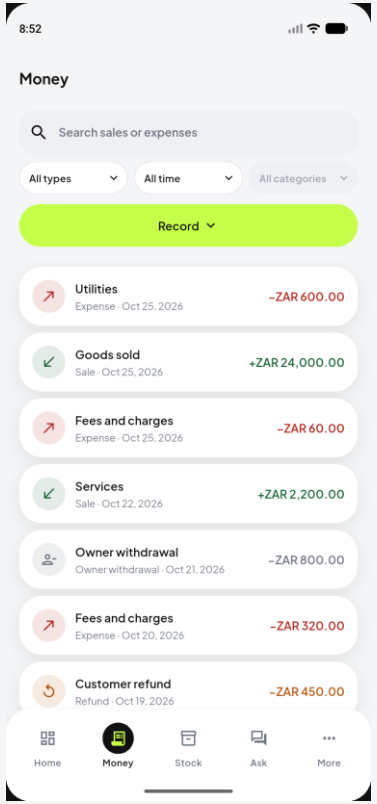
7 / 14 / 30-day outlook subtracts unpaid bills.

Act

Deterministic recommendations plus Ask Ai.

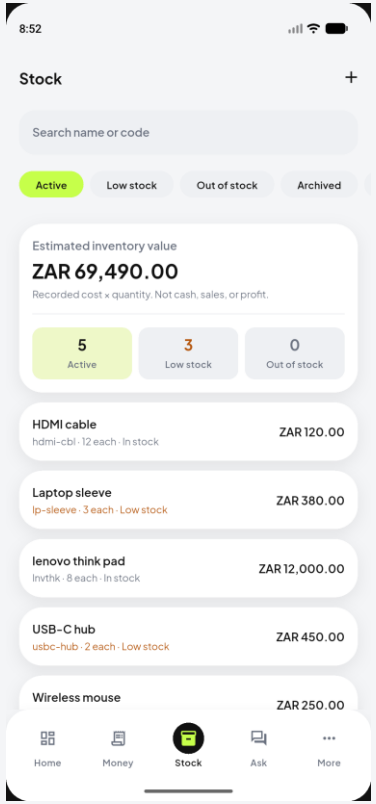


Four screens. Four jobs.



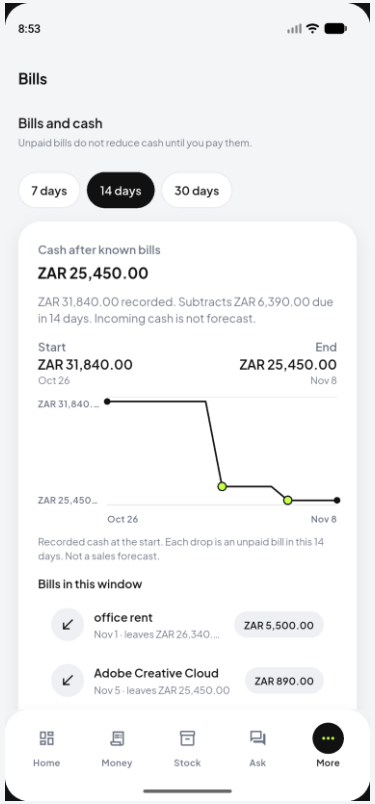
Money

Sales, expenses, refunds and owner movements in one ledger.



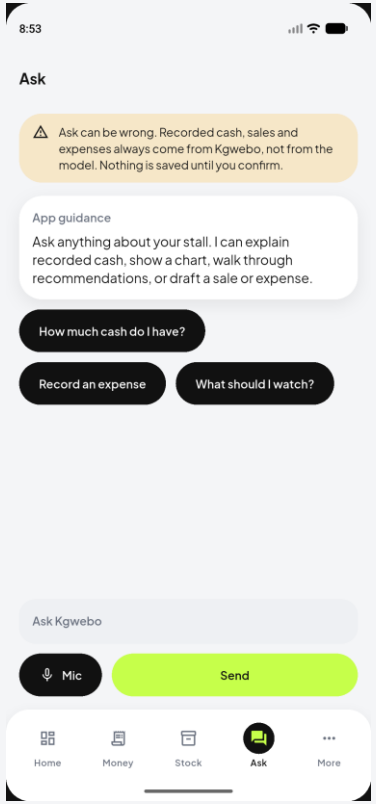
Stock

Catalogue, cost, quantity. Inventory value.



Bills

Unpaid commitments sit in the forecast, not in recorded cash.



Ask Ai

Talk or type. Figures come from Kgwebo. Nothing saves until confirm.

USER STORY · ILLUSTRATIVE DREUTECH

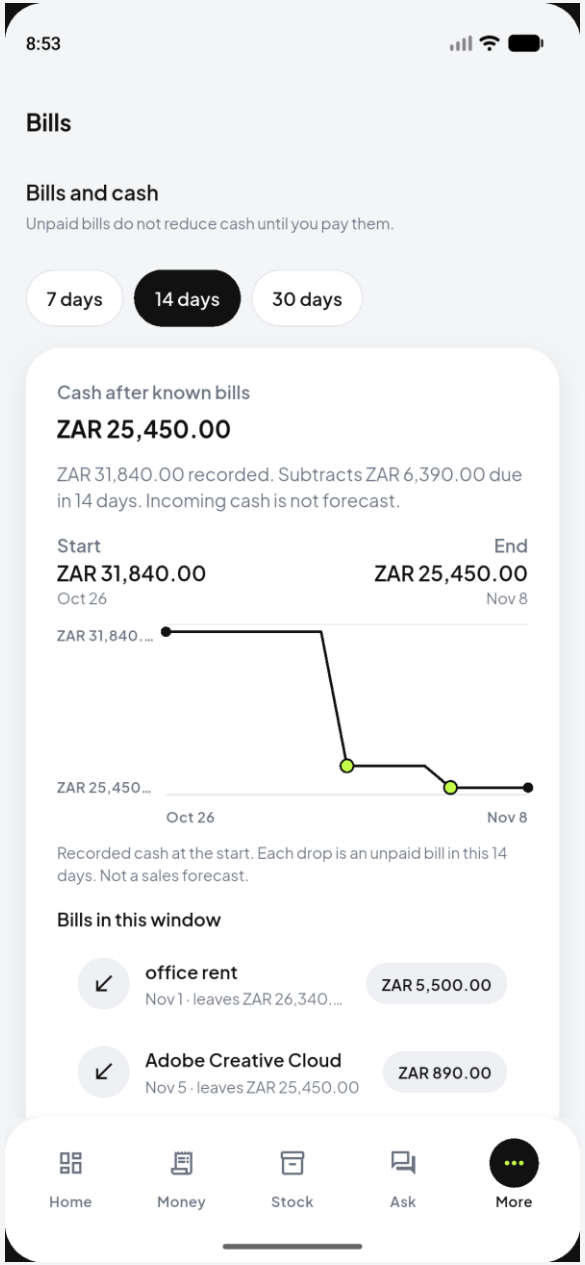
The till looks full.
Rent is due in six days.

Monday 26 October. Dreu runs DreuTech — cables, laptops, and the odd website job. A supplier WhatsApps: ten USB-C hubs, R 4000 cash, today. The till shows R 31,840. Instinct says yes.

What the till does not say: office rent of R 5,500 lands on 1 November. Adobe Creative Cloud of R 890 lands on 5 November. Three products are already low. A client just paid R 450 for a website fix that is not in the book yet.

Recorded cash **R 31,840**	Bills in 14 days **R 6,390**
Supplier offer **R 4,000**	Cash after bills **R 25,450**

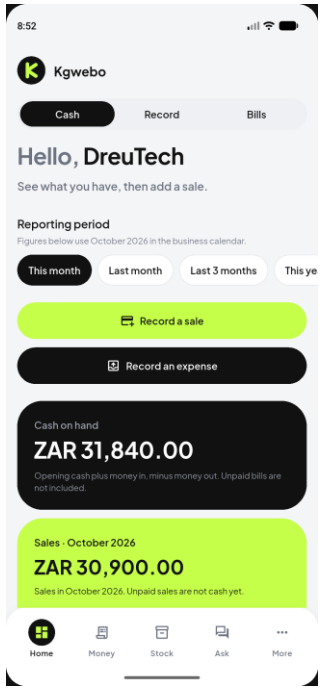
The question is not “is there money in the till?” It is “can Dreu restock and still pay rent on 1st November?”.



Dreu opened Kgwebo and answered it.

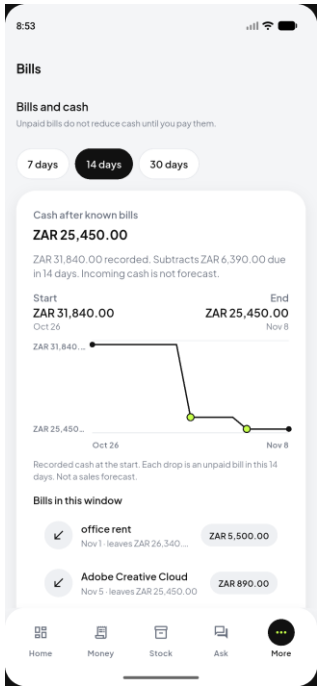
1 See the cash

Home shows R 31,840 recorded cash. Unpaid bills are excluded. This is not free-to-spend money.



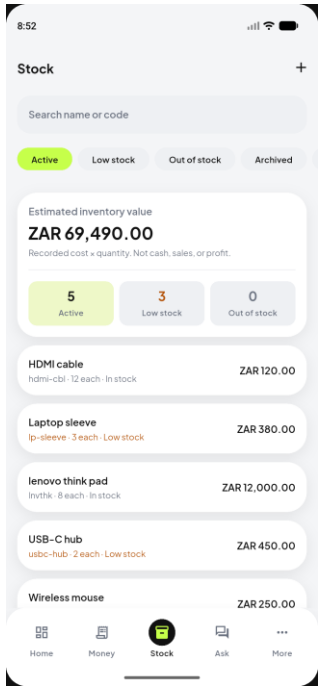
2 Subtract the bills

14-day Bills path: rent 1 Nov, Adobe 5 Nov. After known bills: R 25,450. Incoming cash is not invented.



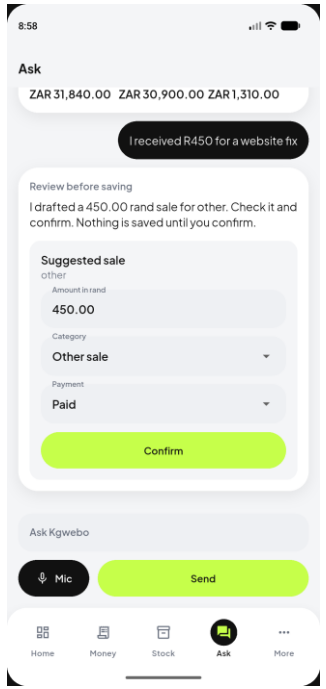
3 Check the shelf

USB-C hub is genuinely low (2 left). Stock on the shelf is R 69,490 — labelled as not cash.



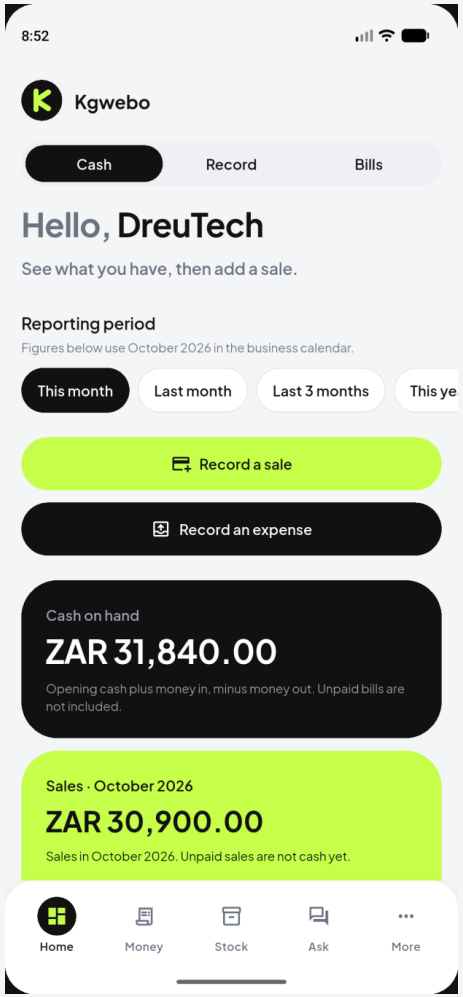
4 Ask Ai, then record

Ask answers cash from records. “I received R 450 for a website fix” becomes a draft. Confirm still waiting.

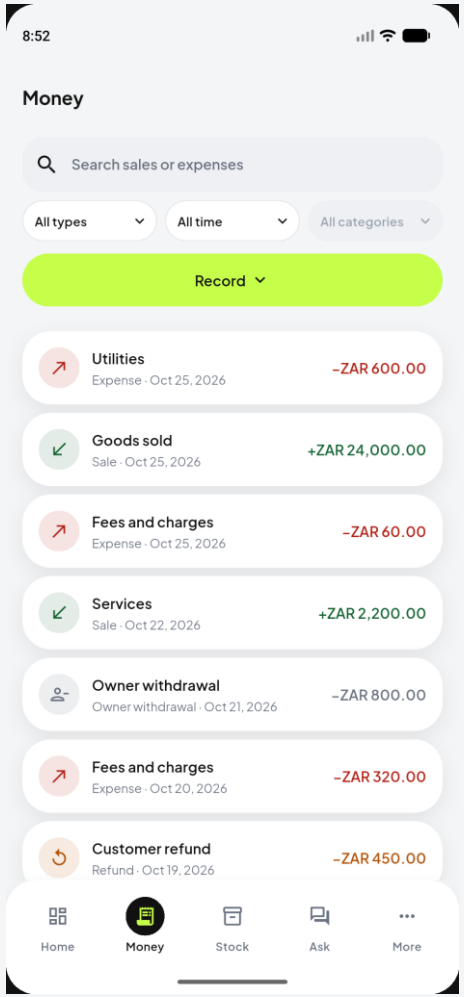


Decision: R 4,000 restock still fits inside R 25,450 after rent and Adobe. Dreu can say yes with eyes open. The R 450 sale is not in cash until Confirm.

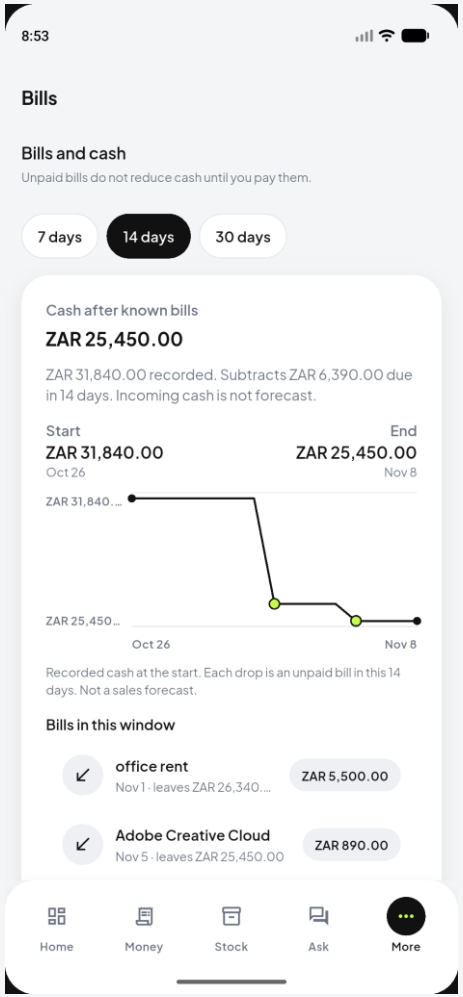
The product, as it runs.



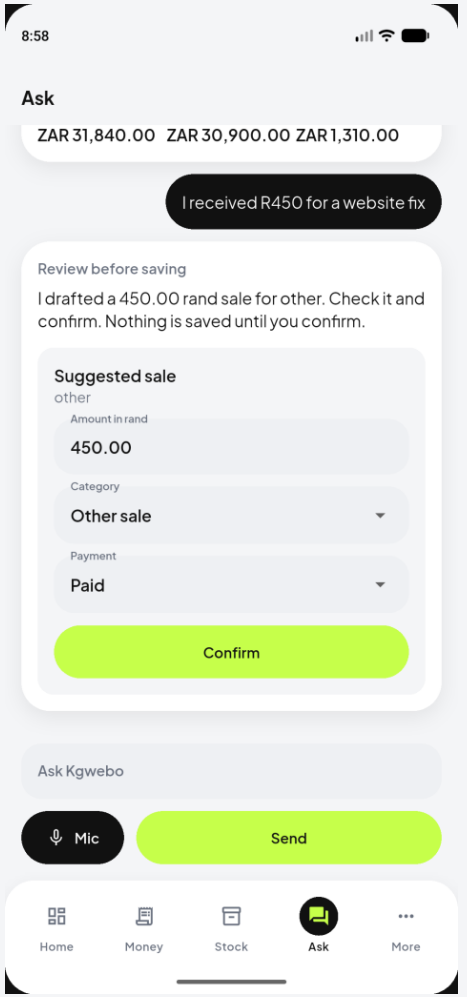
Home · recorded cash first



Money · the actual ledger



Bills · cash after known commitments



Ask Ai · draft, then confirm

INTELLIGENCE

Trustworthy figures first. Then a copilot that cannot quietly change them.

Cash is not sales

Recorded cash = opening cash + paid in − paid out. Dreu's R 24,000 laptop sale is cash only because it is marked paid.

The till is not free cash

R 31,840 is recorded cash. R 6,390 of bills still sit outside it until they are paid.

Bills wait their turn

Unpaid rent does not reduce cash today. The 14-day outlook subtracts it.

Recommendations are rules

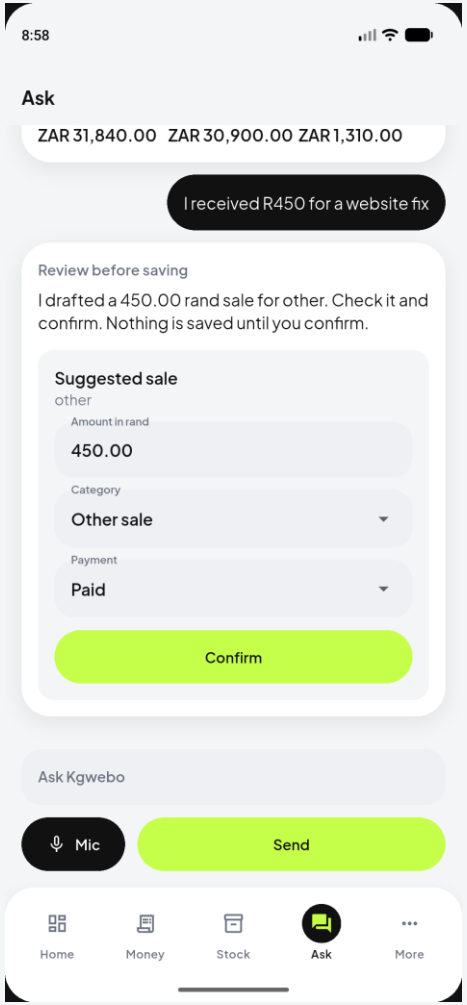
Six evidence-backed rules. Low stock uses catalogue quantities. Copy says estimate, not advice.

Ask Ai is grounded

Ask Ai reads FinancialEngine. Money talk is overwritten with recorded figures. Nothing saves until Confirm.

Stock value is not cash

R 69,490 on the shelf is cost × quantity. Spending it would be spending goods Dreu still needs to sell.



ARCHITECTURE

Built so the phone cannot lie about money.

PRESENTATION

Flutter UI
Android screens. Widgets render values. They never add cash.

GoRouter
Auth guards and onboarding redirects.

Riverpod
One graph of state. Home, Bills, Reports and Ask Ai subscribe to the same summary.

↓

MONEY CORE

FinancialEngine
Pure Dart. Integer minor units.

CashForecast
7 / 14 / 30-day path. Subtracts unpaid bills.

RecommendationRules
Six deterministic rules. Empty stalls stay quiet.

CopilotOrchestrator (Ask Ai)
Classifies intent, grounds replies, never writes the ledger.

↓

ON THE PHONE

Drift cache
Owner-keyed confirmed rows only. Stale cache is labelled, never invented.

Sync outbox
Amount-only creates and new bills. Pending is not cash, sales, or a forecast input.

↓

SUPABASE

Auth
Email/password. JWT on every call. Publishable key only in the APK.

Postgres + FORCE RLS
Ownership frozen on write. A second account cannot read DreuTech.

Inventory RPCs
record_product_sale is atomic. Clients cannot insert stock rows.

kgwebo-Ai
Edge Function. Rate-limited. LiteLLM JSON. No SQL from the model.

One engine feeds Home, Bills, Reports, the PDF and Ask Ai. Isolation lives in Postgres (FORCE RLS, frozen owner_id, SECURITY DEFINER RPCs) — not in the client. There are no bank feeds, payment processors, or silent auto-posts in this build.

The model is a parser. The ledger is the source of truth.

When Dreu records a sale

1. Form validators accept integer minor units. Binary float never holds money.
2. Repository tries Supabase first. Auth JWT. Idempotency key on every create.
3. Amount-only sale or new bill, network fails → durable outbox. Status: pending. FinancialEngine ignores it.
4. Product sale stays online. record_product_sale writes the transaction, line and stock movement in one RPC. Retry cannot double-count.
5. Confirmed row upserts Drift. Home, Bills, Reports, PDF and Ask Ai all recompute from FinancialEngine.

When Dreu asks a question

1. CopilotOrchestrator(Ask Ai) classifies intent from the utterance.
2. A snapshot is read from FinancialEngine — the same R 31,840 Home already shows.
3. JWT call to kgwebo-copilot. Rate-limited Edge Function. LiteLLM returns JSON only.
4. groundCopilotTurn overwrites any invented cash, sales or advice with recorded figures.
5. A draft is shown. Confirm runs CopilotLedgerCommit through the same repository as a typed sale. Cloud down? On-device heuristics, still grounded, still confirm-before-save.

Better habits. Clearer cash. Room to grow.

Daily recording

A stall that captures the R 450 website fix the same morning can see cash move instead of reconstructing the week from chat threads.

Cash that is actually available

Dreu’s restock decision uses R 25,450 after known bills, not the R 31,840 sitting in the till. Money already spoken for stays visible.

Stock discipline

Low-stock notices use saved catalogue quantities. Inventory value is labelled as not cash, so a full shelf does not look like a full till.

A path that stays cheap to run

One Flutter app, one Supabase project, one money engine. Operating cost stays a small backend, not a stack of bank feeds and extra models.

Kgwebo

Every business deserves a clearer picture of its money.

Record what happened. See the cash that is actually free. Act before rent lands.

JUDGE DEMO ACCOUNT

Email: dreu.davids1@gmail.com
Password: password
Stall: DreuTech · illustrative records

